Analysis and Selection of the Force Sensitive Resistors for Gait Characterisation

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***Abstract*— Force Sensitive Resistors (FSRs) are used for kinetic evaluation of human gait by measuring under foot pressures during gait. FSRs from different manufacturers, with variation in morphologies and loading capacities were selected in past research projects. However a functional comparison of different FSRs and options on how to choose them for gait analysis is not established yet. In this work, investigation and comparison of behavioral patterns in measuring normal gait from two major FSR sensors are done. These FSRs had been utilised previously for the purpose of gait sensing. They were placed underneath the heel alternatively to detect heel pressures while wearing footwear with laces and with another footwear possessing straps for harnessing. Both sensors are connected to a miniature low power wireless senor node used for data collection and wireless transmission for off-line analysis of pressure patterns. The results give good indications on capabilities and trade-offs when using the two sensors types.**

***Keywords*— Gait; Force Sensitive Resistors; FSR 406; A401 Flexiforce; Heel pressure**

I. INTRODUCTION

Gait can be defined as a periodic patterned displacement of lower limbs during a walk. It is regarded as fundamental need of an individual to travel small distances and perform daily routine activities. Therefore it has always been the primary goal of clinicians and rehabilitators to study and analyse gait of an individual with walking abnormality before proposing a remedy for it. For this, several methodologies have been adopted, utilising sensors of varying nature to collect temporal and spatial information during gait cycles. The sensing techniques can be classified into three main categories: 1) Gait Kinetics, 2) Gait Kinematics and 3) Physiological Variations.

Each category contains multiple sensing subcategories as discussed in [1-3]. Gait kinetics mainly relates to force measurements under foot while walking. In this context, Kyongchul deployed air bladder and air pressure sensor assembly [4], Hui Yu chose strain gauges [5], M. Saito used conductive rubber sensors [6], while A. Faivre experimented with custom made dynamometric rings [7]. They all highlighted benefits of using these sensors. However, a large number of researchers preferred force sensitive resistor (FSR) for phasic analysis of normal as well as abnormal gaits due to ease of set up and use in real walking scenarios [8, 9].

FSR consists of thin polyester film with conductive piezoresistive substance inside. The force when applied onto films causes suspended particles of substance to come together and make up a conduction path. The resistance of sensor changes inversely with applied load thus behaves as a resistor varying with force [1, 2, 10].

Human gait is characterised by two main phases, the *Stance* and the *Swing* phases. Stance phase is the stage when foot is in contact with ground surface partly or fully while in swing phase it is in the air. The feet execute these phases alternatively to accomplish a gait cycle. Conventionally a gait cycle is considered to start from heel strike on the floor of one foot and to complete at next heel strike of the same foot while going through both phases. Each phase is further partitioned into sub phases as illustrated in Fig 1.

During stance phase, the body weight is supported sequentially in order to displace it forward while transiting through sub phases. It begins with initial contact, known as the 'heel strike', followed by 'loading response' and 'mid stance' where forefoot contacts, laying 'foot flat' on the ground. During this period, the contralateral (other side) leg swings forward for its initial contact causing the weight on ipsilateral (same side) leg to shift from heel towards toe and consequently raising the heel known as 'heel off'. From here 'terminal stance' starts and ends into 'preswing' upon heel strike of the contralateral leg. Preswing terminates with 'toe off', accelerating leg into swing phase, whereas contralateral foot enters its mid stance simultaneously [11].

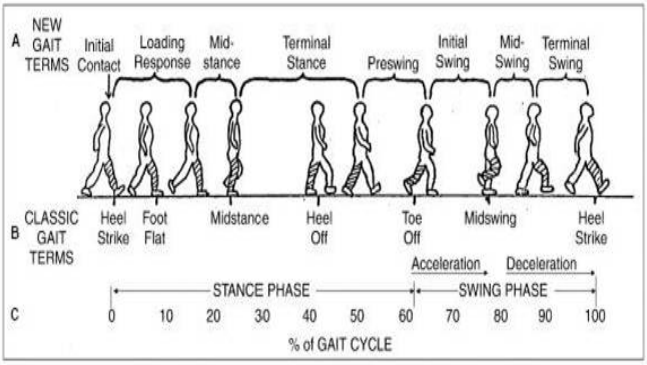

Fig. 1 Gait Cycle, % Phasic Contribution & Terminologies *from[12]*

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Swing phase is sub-divided into 'initial', 'mid' and 'terminal' swing. These sub phases are identified through kinematic (motion describing) differences. Right from start of initial swing, leg accelerates forward under influence of hip and knee flexors and the muscles at ankle joint lift fore foot so as to clear the ground while being in air. This movement at ankle is known as dorsiflexion and is shown in Fig 2. Next, from the middle of mid swing till end of terminal swing, knee extends and the leg decelerates to strike the ground [11].

In this paper, two different force sensitive resistors: FSR 406 and A401 are evaluated while demonstrating gaits of healthy individuals through their loading behavior. Both sensors have been exclusively utilised in prior research for force sensing based definition of gait. In this regard recent work with FSR 406 is present in [13], whereas experimentation with A401 is presented in [14]. However, none of these works discussed the basis of selection for the type of FSR sensor used in their research. In this work, the two FSR types are explored and compared for gait characterisation. For this, the pattern of an individual's gait is pictured in accordance with the pressure changes measured by FSRs on a single foot. The stance and swing phases are represented along with distinct indication of sub-phases according to patented gait definition. Thereafter, performance based selection criteria for FSR types pertaining to individual's gait pattern are developed.

The rest of the paper is organised as follows. Section II describes motivation for this research and discusses the preparations and operational setup for data collection. Section III presents the results of the experiments. In Section IV results are discussed and evaluated in the context of gait analysis. Section V presents the comparison of the two sensors in gait experiments highlighting their advantages and disadvantages, as well as the future directions.

II. METHODOLOGY

A. Concept

Previous works established that heel is a major load bearing part of foot during gait cycle and both stance and swing phase could be distinguished through pressure changes under heel. Moreover, the variations in heel pressures, while executing sub-phases during stance phase, can be an important factor for description of an individual's gait [4, 8, 9].

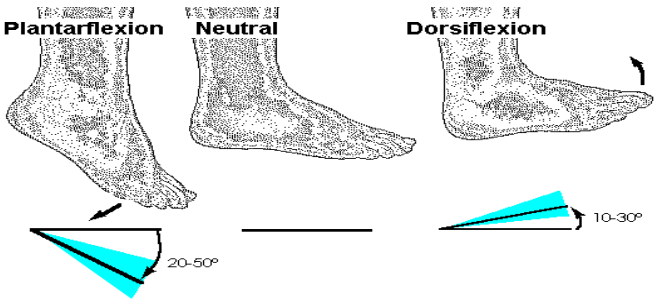


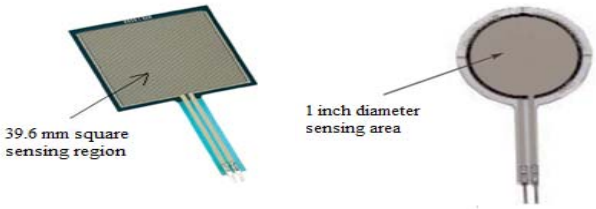
Fig.2 Flexions at ankle joint during gait *reproduced from [15]*

B. Experimental Setup

Two FSRs of different shapes and load bearing characteristics with response time less than 5 micro second were selected. The first type is FSR 406 produced by Interlink Electronics, [16], shown in Fig. 3(a) and the other type is Flexiforce A401 from TekScan [17] (later in text as A401) shown in Fig. 3(b).

As the first step, sensors were selected for experiments after being tested with calibrated weights for the characteristic curve supplied by manufacturers as shown in Fig. 4(a) and 4(b). To ease the comparison, the characteristic curves of both FSRs are combined in a common scale plot as shown in Fig. 5. It is obvious that A401 has a greater functional bandwidth for force changes, ranging from 2 kilograms Force up to 60 kilogram Force and can even go beyond as suggested in curve. On the other hand FSR 406 is more sensitive to forces with lower magnitude, able to distinctively respond to forces as low as 20 grams.

A generic sole was prepared for the purpose of the experiment by marking the region of interest using ink imprints of foot. A pair of pre-sewn pockets for FSR enclosure was pasted one underneath the heel and the other under the great toe. The experiments with different sensor configurations were conducted alternatively to measure the pressures mounted by the heel against ground during gait.



a) **b)**

Fig. 3a) Interlink's FSR 406 *reproduced from[16]*, b) Tekscan's A401 *reproduced from[17]*

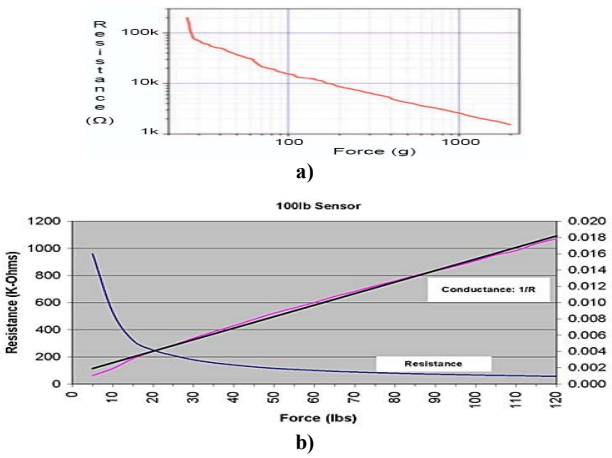


Fig.4 a) Characteristic curve for Force vs Resistance for FSR 406 *from[16],* b) Characteristic curve for Force vs Resistance & Force vs Conductance for A401 *reproduced from[17]*

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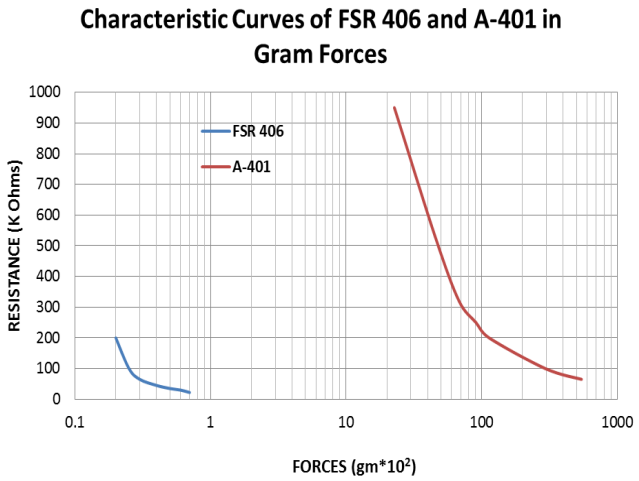


Fig. 5 Characteristic curves for Force vs Resistance for FSR 406 and A401 with Forces in grams and resistance changes in Kohms

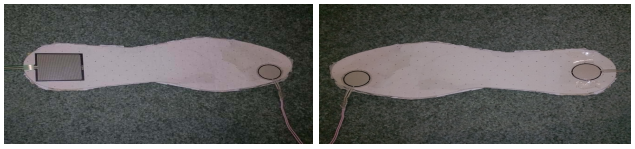
Also, with both FSRs, a separate A401 was placed under the great toe to get better interpretation of phasic change from stance to swing phase as shown in Fig. 6(a) and 6(b). A fixed value resistor is connected in series with the chosen FSR to form a voltage divider circuit. The voltage drop across the fix resistor V_R is fed to an analogue-to-digital converter with the output which is calculated as:

$$V_R = V_S * R / (R + R_{FSR}) \quad (1)$$

where V_S is the source voltage, 3V in our case.

As from the working principle of FSRs, their resistance decreases with increasing load and subsequently the current through FSR increases, which allows major drop of voltage across fixed resistor at the output. The relationship of FSR conductance with applied loading stays approximately linear throughout as reported in [10]. This way the output voltage reflects direct proportionality with the amount of pressure applied during gait activity.

Two participants capable of walking normally, weighing between 65 and 80Kg were asked to walk on a flat surface while the prepared sole was inserted at first in a complete foot enclosing (closed) footwear, tied with laces, and then the experiments were repeated wearing a bit open footwear (strapped) with the straps for fastening. To process the output voltage, a sensor node with Texas Instruments CC430F6137 integrated microcontroller [18] was deployed using two channels of its ADC to collect and process the analogue voltages from the fixed resistors at heel and toe. Data was sampled at 50 Hz, and transferred wirelessly from the sensor node to a workstation for analysis in MatLab. The sensor node was firmly pasted on medial site of the foot as in Fig. 7. Different values of fixed resistors were tested with both FSRs and outputs were observed; finally, the gait patterns generated out while using 1 Kohms with FSR 406 and 100 Kohms with A401 are selected for discussion here.



a) b)

Fig.6 Placement of sensors a) FSR 406 at heel and A401 at toe, b) A 401 on both sites

a) b)

Fig.7 a) Closed footwear setup, b) Strapped footwear setup

III. RESULTS

The outcomes of the experiments turned out to be similar in terms of gait pattern while testing for same type of FSR with both participants. However, tangible differences in gait pattern were commonly observed with both participants upon switching the FSR type. For the sake of brevity, results of only one participant are discussed here to describe commonalities and the varying features with changes of FSR type. From Fig. 8 to 11, it can be seen that with both FSRs, irrespective of shoe type, the load bearing (loading response) indicated as 'L' initiates with heel strike so heel output rises swiftly and reaches maximum until the middle portion of foot's plantar side touches the ground entering into mid stance referred as 'MID ST'. Further, from mid stance to heel off, load shifts from heel to middle foot leading to forefoot. Accordingly, in the plots in Fig. 8-11, this is exhibited by fall of heel output after a sustained maximum level followed by a rise of toe output. As the heel output drops to unloaded values, terminal stance stated as 'TER ST' begins and ends when the toe output nears its maximum where propulsion (pre-swing) abbreviated as 'P' occurs. Next, at toe off, the toe sensor drops to unloaded level indicating transition into swing phase where both heel and toe sensors stay at the unloaded level. A mid value during ascent and descent of the sensor output was selected as border line for sub-phasic transitions.

From Fig. 8 and 9 with A 401 under heel, as the heel strikes, regardless of type of footwear, a significant rise in voltage levels is seen at the output of fixed resistor at heel and shapes in a peak. Later, it begins to drop as the foot goes flat and diminishes when the heel lifts. From here, the output of toe fixed resistor rises referring to terminal stance. Upon toe off, the output of toe resistor decreases to values representing initial swing phase after which no changes in outputs occur until next heel strike. A slight difference in rise pattern of toe is however observed in closed footwear where output from toe increases gradually right from the start of mid stance (Fig. 9).

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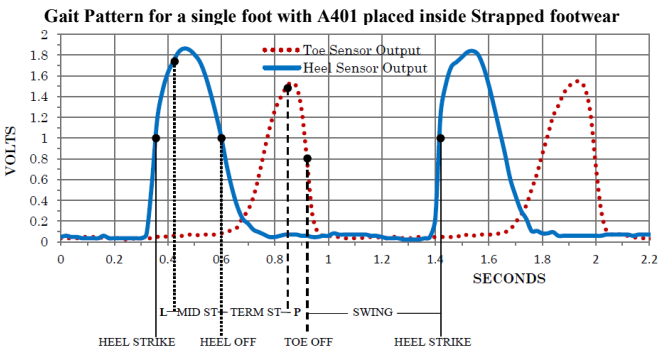

Fig.8 Loading curves with A401 under heel and toe in Strapped footwear

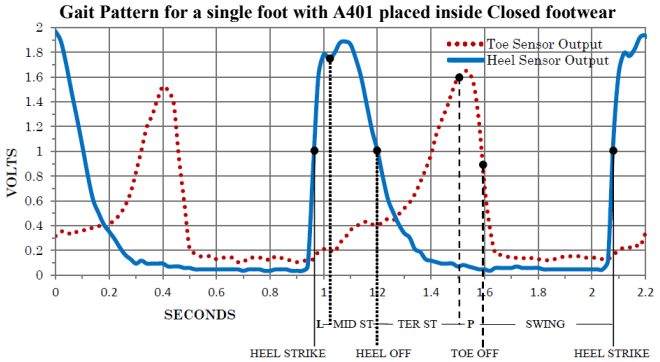

Fig.9 Loading curves with A401 under heel and toe in Closed footwear

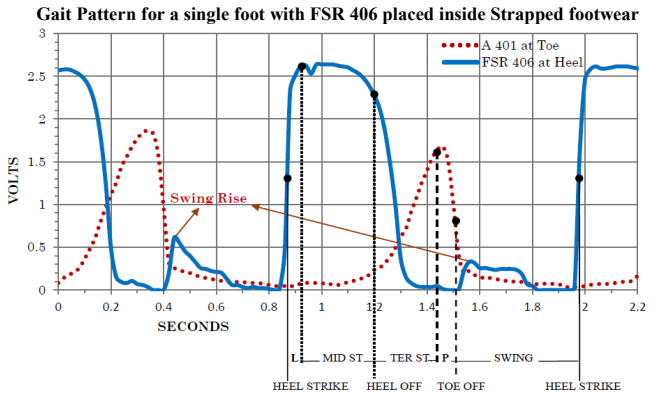

Fig.10 Loading curves with FSR 406 under heel and A401 under toe in Strapped footwear

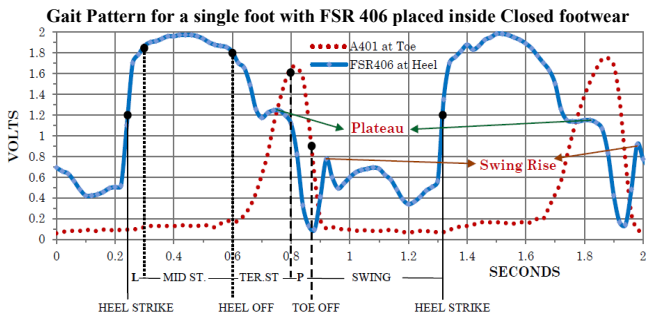

Fig.11 Loading curves with FSR 406 under heel and A401 under toe in Closed footwear

In case of FSR 406, some important differences were noted. After heel strikes the ground, the stay period at maximum level is broadened, which maintains at highest value for a while before heel voltage drops to unloaded value. With the start of swing phase, output from heel rises undesirably to a level higher than unloaded values. The rise persists for the entire swing phase and drops back to the unloaded value just before the next heel strike. This phenomenon was observed with both footwear types and is much higher in closed footwear. It is annotated as 'Swing Rise' in Fig. 10 and 11. Also, in closed footwear, a peculiar level sustenance was observed during the descent of heel output while transiting from mid stance into terminal stance. This observation is marked as 'Plateau' in Fig. 11. Besides aforementioned observations, another discovery while testing for suitable value of fixed resistor with FSR 406 was the rise in magnitudes of these strange formations, which increase directly with series resistor value as shown in Fig. 12 a-c.

IV. DISCUSSIONS

The output reflects varying aspects of two types of FSR, owing to the size, load bearing capacity and enclosure within shoes. Both distinctively describe the commencement of stance phase with substantial rise in voltages from heel side upon Heel Strike first, then indicate end of the stance phase and start of the swing phase with dropped voltages from both heel and toe setups at the instance of Toe Off. However, momentary differences are identified within stance as well as swing phase. These are noticeable in swing phase right after toe off (swing rise); whereas in stance phase, observable near the end of mid stance when the forces at heel are dropped around half (plateau) while the toe forces are increased to a level indicating toe under stressful contact. Beyond this, the terminal stance begins where a characteristic rise in toe pressure occurs that reckons to limb propulsion as depicted in gait plots by A. Faivre in [7]. By analysing the gait patterns in Fig. 8-11, it can be concluded that the loading curves of A401 based setup are much closer to the gait representation in referred work rather than FSR 406 based setup.

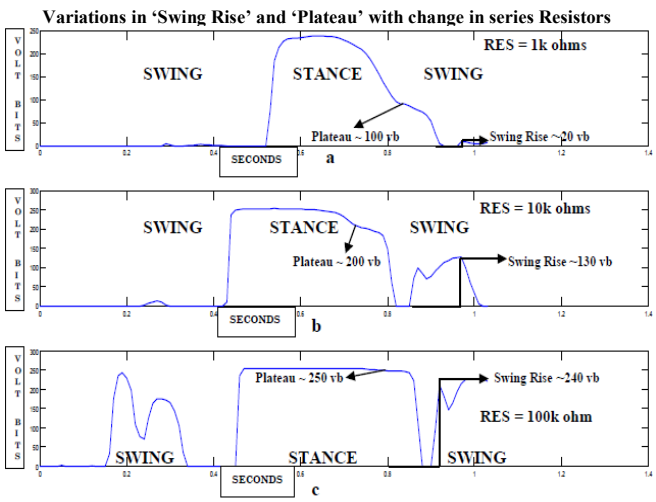

Fig. 12 Variations in Swing Rise and Plateau formations with FSR 406 in Closed footwear a) with 1k ohm b) with 10k ohm c) with 100k ohm

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A) Effects of Footwear Enclosure

Increasing toe forces were observed right from mid stance while wearing closed footwear as shown in Fig. 9 and 11. This is caused due to constrained space for forefoot positioning in closed footwear. Due to this, the toe sensor gets stressed immediately after foot goes flat as toe comes into an earlier contact with the sensor.

B) Influence of Sensor's Surface Area

The 'plateau' formation with FSR 406 in closed footwear referred in Fig. 11, suggests the existence of sustained pressure by heel onto the sensor, referring to a persistent contact of heel with sensor while transiting load from heel to forefoot. In the strapped footwear, heel displaces more freely and depresses the sensor in a smoother pattern and therefore 'plateau' is not observed with strapped footwear. Also its absence in cases of A401 highlights that the occurrence probability of plateau goes with sensor's surface area as well. Although with greater surface area of FSR 406 one has to deal with such unnecessary formations, yet it adds up to the benefits by increasing possibilities to detect heel strikes in events where heel might be striking unevenly and establishing contact with the ground on the sides rather than the middle.

C) Working Range and Sensitivity of Sensor

Fig. 5 suggests FSR 406 is much more sensitive than A401 for smaller forces, however not able to embrace larger loads as A401 does. Therefore difference in loading behavior of two sensors owes its existence to differences in sensitivity characteristics between them.

It's observed that unlike A401, FSR 406 doesn't display a peak formation during loading response and mid stance when heel loads shift from striking point to the heel off point. This is for the very reason that the forces are typically quite large in any point in time during transition from heel strike till heel off and FSR 406 is unable to distinguish them in terms of resistance. Hence a sustained level is seen right from heel strike till heel off while using FSR 406.

Next, in start of swing phase, 'swing rise' is observed while walking with FSR 406 as shown in Fig. 10 and 11. Swing rise occurs due to ankle dorsiflexion during 'initial swing' which in turn causes minor depression by heel onto sensor that is falsely interpreted as if heel loading in ground contact. That is why combining higher values of series resistances with FSR 406 is not recommended as this would increase magnitude of swing rise as well as of plateau. Fig. 12 elaborates this issue graphically, where resulting output voltages are shown in bits scale. Starting from Fig. 12(a), it's observed that a 1 Kohm fixed resistor results in a very little swing rise of about 20 bits which is almost negligible whereas plateau is showing rise of 100 bits. However a 10 times increase in series resistor lifts swing rise at 130 bits that is equal to half of the magnitude of stance phase value and plateau nears the mid stance values reaching to 200 bits as shown in Fig. 12(b). The worst case is observed in Fig.12(c), where a series resistance of 100 Kohms with FSR 406 gives a boost to swing rise and plateau equalising almost maximum loading values at heel. Here the mid stance fall is merged with terminal stance rise due to plateau while the swing rise portrays a heel strike. This situation could be disastrous if an algorithm is distinguishing phases and sub phases of gait by monitoring changes in heel loading.

On the other hand A401 can bear much higher loads as compared to FSR 406. Normally A401 can measure 100 lbs and with certain modifications in driving circuit, it can measure up to 7000 lbs [17]. Beside this, as suggested from Fig. 5, A401 exhibit noticeable variations in resistance for larger magnitude of forces, thus for smaller forces such as those mounted by ankle dorsiflexion during swing, it appears to be insensitive and so does not vary its resistance at all. Hence swing rise is not produced while using A401.

D) The power consumption

The square shaped FSR 406 showed undesirably high sensitivity to forces and quickly dropped to low resistive values with rising forces. Therefore, a low value resistance like 1 Kohm was selected to get distinct patterns for stance and swing phase though this low resistance configuration led to rapid discharge of batteries due to higher current drain through the voltage divider.

The round sensor A401 showed lower sensitivity to foot movements during swing phase with distinguished smooth patterns for values as large as 330 Kohms. However 100 Kohms is used for optimum output voltages. Thus A401 has edge in terms of power consumption of the system, as well.

From above mentioned discoveries, selection criteria are summarized in following table based on advantages of both FSRs with regards to gait scenario.

TABLE I. BENEFITS OF FSR 406 OVER A401

	A401	FSR 406
1.	***Enhanced probability of Heel strike detection***	
	Due to smaller contact area (0.8 in^2 approx. from 1 inch dia.), A401 might miss out heel detection in situations when heel does not strike the ground at the center. Such a case could be seen when surface is not even or foot is unexpectedly landed with excessive inversion or eversion.	FSR 406 can benefit from larger surface area (2.6in^2 approx. from 39.6mm each side) of contact while walking on uneven ground where foot might strike on the side, rather than middle of heel. Thus the probability of heel strike detection is higher than A401 in case of non-flat surface.
2.	***Suitable for low loading activities***	
	A401, because of its lower sensitivity for smaller forces (2kgF minimum force to vary resistance noticeably), is not envisaged to work well for low loading activities like gait analysis in pediatrics (children) or individual experiencing some foot loading problem.	FSR 406 shows increased sensitivity for lower loads (20 gmF sufficient force to produce resistance change). Thus it could be useful for gait analysis in pediatrics and in cases where individuals are unable to support body weight on the heel completely due to trauma or any other gait related suffering.
3.	***Economical***	
	A401 is quite expensive than FSR 406. 1 A401 is about $24NZD in a pack of 4 sensors.	FSR 406 is much cheaper even less than half of A401 price. 1 FSR 406 is worth 10NZ$ approximately.

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TABLE II. BENEFITS OF A401 OVER FSR 406

	A401	FSR 406
1.	***Precise representation of phases***	
	A401 offers M Ohms resistance to force values below 2 Kilogram. This feature nullifies the effects of false loading at heel during dorsiflexion in swing phase (no swing rise). Also by virtue of smaller surface area exposed to loads (a circular perimeter of 3.1 inches approx.) the heel forces distribute symmetrically while shifting from rear to fore foot during transition of mid stance into terminal stance. (no plateau)	FSR 406 being sensitive to as low as 20 grams force or even lesser, detects heel depression due to dorsiflexion in swing phase and output as a true compressive loading against ground (swing rise). Unlike A401, it offers larger contact area under heel. Quantitatively a square perimeter of 158.4 mm (6.2 inches approx.). So progression of heel forces show abruptness during transition from rear to fore foot for midstance into terminal stance. (plateau formed)
2.	***Suitability for normal to high loading activities***	
	A401 is capable of bearing greater pressures. Typically for a loading of 100 lbs, it offers resistance around 100 Kohms as shown in the curve in figure 4b. This highlights its usefulness in analysis of extreme loading activities like running, jumping, walking down a steep slope etc.	FSR 406 reduces to resistance values under 3 Kohms while sustaining just about 1 Kilogram force (2.2 lbs only) as suggested by the curve in figure 4a. Due to this feature it doesn't appear to suit for high loading activities like running, jumping, walking down a steep slope etc.
3.	***Power effectiveness***	
	A401 promotes longer battery life by offering high ohmic pathway to reduce the current drain. For example to drop 1 V in this experiment, sensor would be at 50 Kohm (series resistance 100 Kohm and battery 3V). Hence a current of 20 µA would flow through the resistive path.	FSR 406 has to be worked out with smaller fixed resistors this would enforce large drain of current subsequently reducing battery life. With FSR 406, 1 V drop would occur when it's offering 500 ohms (series resistance 1Kohm and 3V supply). Eventually a current drain of 2mA would result.

V. CONCLUSIONS & FUTURE DIRECTIONS

In this paper, two force sensitive resistors (A401 and FSR 406) used in past gait research, were compared while being tested in same conditions. Based on evaluation, selection criteria are established, which according to authors' knowledge, were never proposed before. The criteria founded in our research could be beneficial for future FSR based gait investigations, providing guidance for choosing suitable FSR for specifics of the gait conditions.

With both sensors, gait phases were vividly detected while participants walked in real walking situation on level surfaces. Further from analysis, it is deduced that A401 by virtue of its robust load withstanding capacity, can be advantageous in cases of adult gait as well as for some extreme gait activities in which large pressures are mounted by the foot. Also with lower sensitivity profile of A401 for low magnitude forces, one can develop reliable algorithm for gait phases' detection while using higher values of series resistances with it, which also offers long period of battery usage in experiments.

So far the tests were run on adults only as per permission granted through ethics approval and the gait of children or participants exerting lower pressures on heel due to a walking problem or any other reason is required to be investigated for further evaluation. Testing on non-flat surfaces and with terrain changes like stairs, inclined scenarios are future targets which would surely provide additional supportive information for the selection criteria. More quantitative measures have to be extracted that indicate possibility of defining algorithms for characterisation of gait phases (and sub-phases). Also power consumption with both types of sensors is to be quantified with more definite measurements.

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